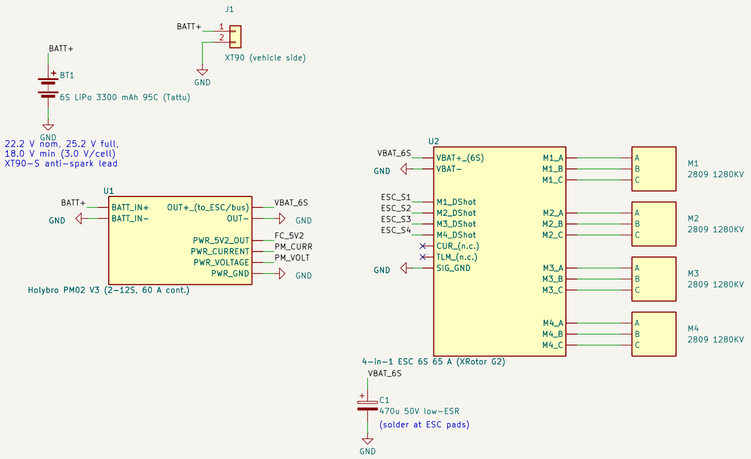
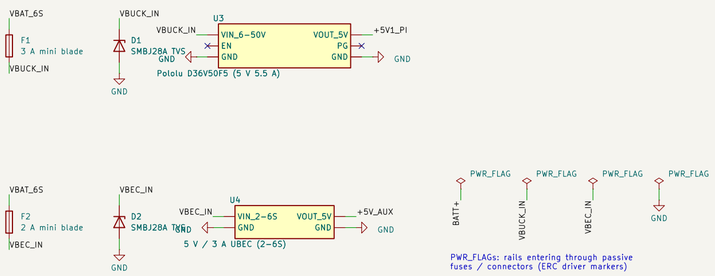
SURANG avionics — power distribution and signal interconnect | 6S 3300 mAh | Pixhawk 6C Mini + Raspberry Pi 5

P. 6S power: battery, power module, ESC, protected 5 V rails



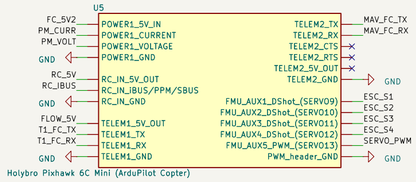
Protected 5 V branches (fuse + TVS crowbar: reversed battery -> TVS conducts -> fuse opens)

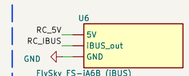


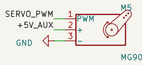
Motor order (ArduPilot X):
M1 front-R CCW, M2 rear-L CCW,
M3 front-L CW, M4 rear-R CW
7x4x3 tri-blade props

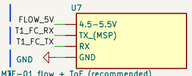
Buck EN left open (internal pull-up = on); PG unused. Pi fed by a 10 cm 20 AWG USB-C pigtail (no USB-PD):
set usb_max_current_enable=1. ESC branch is protected only by the keyed XT90 — never hot-plug.

F. Flight controller, RC, cassette servo



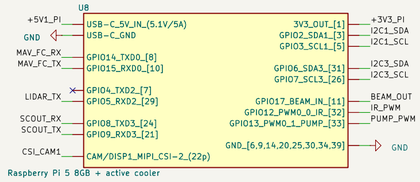


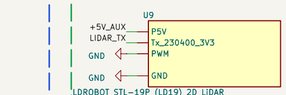


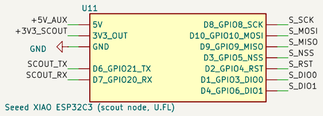


TELEM2 <-> Pi UART0: MAVLink2 921600 8N1, no flow control (BRD_SER2_RTSCTS=0). VCC pin not connected.
ESC: DShot600 on FMU AUX1-4 (SERVO9-12; IO MAIN outputs do not do DShot). Servo: AUX5 = SERVO13 PWM.
RC: FS-iA6B iBUS -> RC IN (if not decoded, move to a spare UART with SERIALx_PROTOCOL=23).
TELEM1: optical flow + ToF (MSP, 115200) for GPS-free Loiter in the gallery.
Failsafes: RC + GCS loss -> brake/hold -> land; battery 2-stage (BATT_LOW/CRT); props idle when perched.

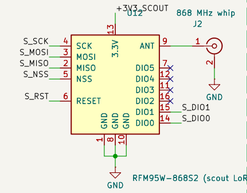
C. Companion computer, LiDAR, camera, scout radio node





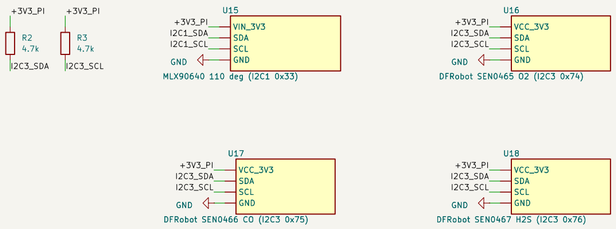






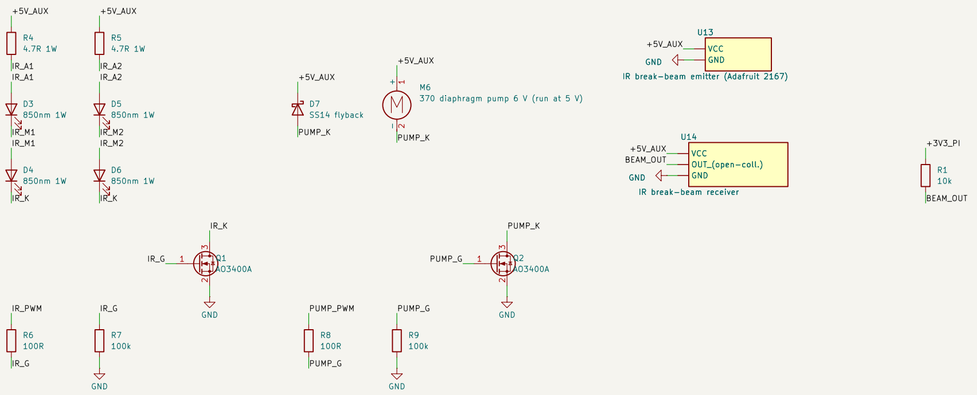
UART0 GPIO14/15 = MAVLink (FC TELEM2)
UART2 GPIO5 RX = LiDAR 230400 (dtoverlay=uart2-pi5)
UART3 GPIO8/9 = scout node 921600 (uart3-pi5)
LiDAR PWM tied to GND = internal 10 Hz scan
IMX327: 22->15-pin FPC, 2-lane, dtoverlay=imx290
Scout XIAO 3V3 LDO feeds RFM95W (peak ~0.47 A)

S. I2C sensor buses (3.3 V, Pi-hosted)



I2C ADDRESS MAP
I2C1 (GPIO2 SDA / GPIO3 SCL), 400 kHz-1 MHz, Pi on-board fixed pull-ups:
0x33 MLX90640 thermal array (alone on the bus so it can run FM+)
I2C3 (GPIO6 SDA / GPIO7 SCL, dtoverlay=i2c3-pi5), 100 kHz, R2/R3 4.7k pull-ups:
0x74 SEN0465 O2 (DIP A0=0 A1=0)
0x75 SEN0466 CO (DIP A0=1 A1=0)
0x76 SEN0467 H2S (DIP A0=0 A1=1)
0x77 DFRobot factory default - left unused so a fresh sensor cannot collide
Gas boards: SEL DIP = 0 (I2C), powered from 3V3 so SDA/SCL never exceed 3.3 V.
CH4 NDIR (Ambetronics OEM) is planned on UART/RS-485 via an adapter, not on these buses.

L. Switched loads on +5V_AUX: 850 nm illuminators, sample pump, break-beam



IR: 2 strings x (2 x 850 nm 1 W LED + 4.7R) = 2 x ~0.37 A from 5 V (Vf ~1.6 V each); Q1 low-side, PWM dimmable on GPIO12.
Pump: 370-class diaphragm pump, ~0.3 A at 5 V, low-side Q2 + SS14 flyback; GPIO13 PWM sets sample flow.
Gate network: 100R series (edge/ringing) + 100k pull-down = load OFF while the Pi boots or is unpowered.
AO3400A: logic-level (Vgs(th) <= 1.45 V, Rds(on) ~ 40-50 mOhm at 2.5-4.5 V), 30 V / 5.7 A.
Break-beam receiver is open-collector: pulled up to the Pi 3V3 (R1) so 5 V never reaches GPIO17.

N. Wiring rules

All logic is 3.3 V: FC UARTs, Pi GPIO, LiDAR Tx, ESP32-C3. No level shifters.
Single-point ground at the PM; signal harnesses carry their own GND wire.

Wire gauges: battery/PM/ESC 12 AWG;
fused branches 20 AWG; signals 26-28 AWG.

Current envelope: hover ~25 A (calc),
PM02 60 A cont./100 A burst -> cap total
at 80 A with MOT_BAT_CURR_MAX (amber).

Radios: 2.4 GHz RC + ESP-NOW-LR share
the band - keep RC on high channel hop
and the scout on Wi-Fi ch 1/6/11 plan.
LoRa 865-867 MHz, PA_BOOST <= +17 dBm.

Not built; values are design targets.

Generated by electronics/scripts/build_avionics.py
Modules drawn at connector level: harness not yet built
Gas-free mock-gallery prototype: not intrinsically safe
Vishwakarma Awards 2026-27 Stage 2 design baseline
Team Palanteen, TSEC Mumbai - VK26-AU-PAI-0065

Sheet: /
File: surang-avionics.kicad_sch

Title: SURANG avionics — interconnect + power distribution

Size: A2	Date: 2026-10-22	**Rev: S2-A**
KiCad E.D.A. 10.0.7		Id: 1/1